Deconstructing Greatness

The Big Three GOAT Engine

An end-to-end tennis analytics pipeline, spatial feature engineering framework, machine learning engine, and four-page interactive Tableau Public dashboard.

Proposal at a glance	Planning baseline
Research cohort	Roger Federer, Rafael Nadal, and Novak Djokovic
Analysis window	2010–2022, inclusive
Delivery approach	Modular Python pipeline • SQLite • PCA • XGBoost • Tableau Public
Proposed duration	10 weeks; approximately 100–140 hours for one contributor
Document status	Planning proposal, version 1.0 • 29 September 2026
Academic level	BSc Data Science portfolio / capstone project

Purpose

Build a reproducible, evidence-led comparison of three exceptional tennis careers within a defined era. Translate performance, pressure resistance, surface versatility, and tactical behavior into transparent metrics that viewers can explore rather than treating a single ranking as an unquestionable verdict.

Planning note

This proposal expands the supplied technical specification into an implementation plan. Proposed scores, model performance, sample sizes, and player-specific narratives are hypotheses or design targets until measured. The core architecture and seven checkpoints are retained, with feasibility gates added where the original specification depends on unverified data availability.

Executive summary & research scope

Problem and opportunity

The GOAT debate combines achievements, playstyle, surface specialization, longevity, and subjective preferences. A dashboard can make these tradeoffs visible, but only if its coverage, metric definitions, and uncertainty are explicit. The project will connect descriptive comparisons to reproducible data processing and a separately evaluated predictive task.

Research questions

1. How do the Big Three compare on dominance, pressure performance, and surface consistency during 2010–2022?
2. Which measurable patterns distinguish their serving, returning, and rally behavior?
3. How sensitive is a composite GOAT ranking to user-selected weights and incomplete charting coverage?
4. Can information available before a service game predict whether the server holds, with performance above a simple baseline?

In scope	Outside the core scope
Men’s matches involving at least one Big Three player; hard, clay, and grass surfaces.	A definitive all-era or all-player GOAT conclusion.
Historical match summaries, charted point sequences, game-level predictions, and four dashboard pages.	Live scoring, online model inference, or a production web application.
PCA of eligible player-match observations; broader ATP comparison cohort if explicitly acquired.	Assuming three players alone provide a representative all-ATP archetype map.
Zone-based placement visuals with clear provenance.	Claiming measured tracking trajectories from coded charting data.

Expected outcomes

A documented Python and SQLite pipeline; validated match, point, shot, and game tables; a feature dictionary; exploratory notebook; PCA report; baseline and XGBoost evaluation report; bootstrapped uncertainty outputs; three governed Tableau extracts; and a four-page Tableau Public workbook with a concise presentation narrative.

Success principle

A strong result explains what the data supports, where the sample is incomplete, and how conclusions change under alternative assumptions. Publishing accurate limitations is part of project success even if the final model or ranking is less dramatic than expected.

Data strategy & feasibility gates

Primary data sources

The Match Charting Project repository provides men’s match metadata and decade-specific point files, alongside aggregate statistics and a data dictionary [1]. Use charting-m-matches.csv, charting-m-points-2010s.csv, and charting-m-points-2020s.csv as the initial charting inputs. Freeze the downloaded version and record its commit, retrieval date, and file hashes.

For complete-era match outcomes and match statistics, evaluate the complementary tennis_atp repository [2]. Treat it as an additional planned source, not an already integrated dataset. Build a crosswalk using date, tournament, round, and canonical player identities; review ambiguous joins and retain unmatched records.

Gate	Required evidence	Decision
G1: coverage	Match counts by player, year, surface, and format; charted share of match cohort.	Retain full-match and charted-match denominators separately.
G2: fields	Actual CSV headers, coded-shot documentation, score state, game boundaries, and available metrics.	Confirm target and features before locking model design.
G3: spatial fidelity	Evidence for observed coordinates versus direction/depth categories.	Use labeled zone centroids when exact locations are absent.
G4: optional variables	Serve-speed observations and dated racquet specifications with provenance.	Exclude unsupported innovation components or use approved proxies.

Acquisition and ingestion

Inspect metadata first; standardize player aliases and dates; identify eligible match IDs; stream point files in 50,000-row chunks; filter before insertion; and use transactions with parameterized SQLite writes. Track raw, retained, rejected, and duplicate row counts. Re-running the same inputs must not create duplicate observations.

Coverage and licensing

The charting sample should not be assumed to represent every match in the era. Publish coverage alongside analytical results. Review source licenses and attribution requirements before redistributing extracts or publishing the workbook; record the license applicable to each frozen source. Unknown licensing or insufficient sample support blocks that component, not the entire project.

Architecture & database contract

System workflow

Layer	Responsibility	Artifact
Source snapshots	Download, provenance, checksum, schema inspection.	Immutable raw files + manifest
Ingestion / cleaning	Filter cohort, parse codes, validate identities and dates.	goat_master.db + quality log
Feature engineering	Player-match, game-start, surface and era aggregates.	Versioned feature tables
Analytics / modeling	EDA, PCA, baseline, XGBoost and bootstrap analysis.	Model artifact + evaluation outputs
Exports / presentation	Governed CSVs, offline weight calculations, interactive views.	Tableau workbook + documentation

Canonical schema

big_three_matches_2010_2022: match_id TEXT primary key; match_date TEXT (ISO date); match_year INTEGER; player1, player2, surface, tourney_name, winner TEXT; best_of INTEGER; source_id TEXT. Preserve charting player order. In outcome-based data, map winner_name and loser_name through a documented adapter and populate winner separately.

points: composite key (match_id, point_id); set_id, game_id, server, returner, score_before, point_winner, raw_first_serve_code, raw_second_serve_code, parse_status. This intermediate table is necessary to reconstruct service games reliably.

big_three_spatial_shots: shot_id primary key; match_id foreign key; point_id; shot_sequence; player_name; opponent_name; surface; shot_type; court_zone; shot_x, shot_y REAL; coordinate_provenance; render_x, render_y REAL; is_synthetic_jitter; is_winner; is_unforced_error. Unique source keys prevent duplicated shot expansion.

model_metrics_ci: run_id, execution_date, model_version, target_name, split_strategy, metric_name, estimate, ci_95_lower, ci_95_upper, n_matches, n_games, bootstrap_seed. Keep best_cv_roc_auc as a tuning diagnostic rather than confusing it with held-out performance.

Integrity rules

Enable PRAGMA foreign_keys=ON; index match_id, match_date, player and surface; reject orphan points and shots; use CHECK constraints for binary flags and supported surfaces. Unknown outcomes remain NULL with an explicit reason. Store provenance and quality metadata outside the display extracts as well.

Cleaning & spatial engineering

Parsing and missing values

Shot strings are structured codes. Parse them using the source dictionary before extracting direction, shot type, or outcome. Apply pd.to_numeric(..., errors="coerce") to fields that are genuinely numeric after parsing; never apply numeric conversion to an entire shot string as a substitute for decoding it. Log newly coerced missing values and preserve the original code for review.

Validate chronology, participant identities, winners, point totals, game reconstruction and surface aliases. Convert known binary outcomes to 0/1 while retaining unknowns as nullable integers. Do not convert missing winners or errors to zero. Export a quality report with completeness, duplicates, parsing failures and coverage by analytical subgroup.

Imputation policy

For model inputs, fit imputation within each training split using prior player-surface history only. Rolling seasonal medians must use observations available before the prediction date, with a training-derived fallback when history is insufficient. Add missingness flags. Descriptive rates should expose their observed denominators rather than silently filling unavailable events.

Court coordinate convention

Use a 78 ft × 27 ft singles-court display. Let X run along court length with the net at 0, server baseline at −39 and receiver baseline at +39; Y spans −13.5 to +13.5. Reorient ends consistently and record the transform. Distinguish serve destination from rally destination and separate ad/deuce court where the source supports it.

Direction/depth codes may support zones rather than measured XY locations. In that case, assign documented representative zone centroids and mark coordinate_provenance="zone_centroid". Reserve "observed" for genuinely measured coordinates. Heatmaps describe charted placement categories; they must not imply precise ball tracking.

Visualization only: render_x = shot_x + Normal(0, 0.5^2); render_y = shot_y + Normal(0, 0.5^2). Units: feet.

Generate jitter with a fixed seed, use it only in render fields, and flag is_synthetic_jitter=1. Keep analytical coordinates unchanged. Respect zone and court bounds so jitter does not fabricate out-of-court shots. Where unsupported, prefer discrete zone tiles over smooth density maps.

Surface and format normalization

Standardize Hard, Clay and Grass; explicitly report excluded or unresolved surfaces. Compare rate-based measures using relevant point, game, or set denominators. Stratify best-of-three and best-of-five where endurance conclusions depend on match format; per-set scaling alone does not remove every format difference.

Metric definitions & GOAT scoring

All rates below use proportions from 0 to 1, with numerator, denominator, source coverage and aggregation grain retained. Composite weights are design choices and require sensitivity analysis.

Dominance Ratio (DR)

$$DR = \text{Return points won rate} / \max(1 - \text{Service points won rate}, 0.00001)$$

Aggregate underlying counts before forming the rate. Flag a near-zero denominator and test the effect of extreme DR values. Do not interpret epsilon-protected values as automatically stable estimates.

Clutch Index (CI)

$$CI = 0.5 \times \text{BP saved} / \text{BP faced} + 0.3 \times \text{Tiebreaks won} / \text{Tiebreaks played} + 0.2 \times \text{Deciding sets won} / \text{Deciding sets played}$$

When a denominator is zero, the component is undefined. For the headline comparison, require all components and display sample sizes. A secondary partial index may renormalize available weights but must be separately labeled. Test uncertainty using match-level resampling.

Surface Adaptability Index (SAI)

$$SAI = 1 - \sqrt{\frac{1}{3} \times \Sigma_{\text{surfaces}} (\text{WinRate}_{\text{surface}} - \text{WinRate}_{\text{overall}})^2}$$

This is a surface-consistency index relative to overall win rate, not the standard deviation around the unweighted surface mean unless those centers coincide. High consistency can coexist with mediocre results. Always show overall and surface win rates alongside SAI; require evidence for all three surfaces.

Tactical Innovation Score: exploratory extension

$$Innovation = 100 \times [0.3 \times \Delta\text{NetFreq} + 0.3 \times \Delta\text{ServeSpeedVar} + 0.2 \times \Delta\text{DropShotFreq} + 0.2 \times \Delta\text{RacquetSpecShift}]$$

Retain the original 0.3/0.3/0.2/0.2 weights, but define each Δ on a comparable 0–1 scale before claiming a 0–100 score. Compare 2010–2014, 2015–2018 and 2019–2022. Proposed implementation: average absolute change between adjacent era values, normalized by preregistered bounds and clipped to [0,1]. This measures tactical change, not proven improvement. Unsupported speed or equipment variables prevent publication of the original composite; publish available components instead.

Preference-driven GOAT simulator

Normalize eligible dominance, clutch, overall performance and versatility metrics using a documented reference population. Score = Σ(wk × normalized metric k), with nonnegative weights summing to 1. Recompute rankings and bootstrap intervals under each weight setting. The supplied 94%, 91% and 97% innovation values are design placeholders and must be replaced by calculated results or omitted.

EDA & interpretable PCA

Exploratory analysis plan

Begin with cohort size, charted coverage, missingness, distribution plots, surface splits and year-by-year rates. Compare player profiles using identical windows and denominators. Estimate peak rolling three-year win percentages; publish the window definition and minimum match count. Grand Slam conversion requires an explicit denominator, such as titles per entered tournament, rather than an undefined percentage.

Candidate 15-feature dictionary

Feature group	Candidate continuous features
Serve (4)	First-serve in rate; first-serve points won; second-serve points won; ace rate.
Return (3)	Return points won; first-serve return points won; second-serve return points won.
Pressure (2)	Break-point save rate; break-point conversion rate.
Rally / attack (6)	Net-approach frequency; net points won; mean rally length; long-rally share; winner rate; unforced-error rate.

The feature list is provisional until the field audit. Define long-rally threshold, net-approach denominator and winner/error denominator in the feature dictionary. Remove unavailable or unstable features rather than creating unsupported values merely to reach 15. Avoid double-counting strongly redundant inputs without a sensitivity check.

PCA specification

Use player-match observations with sufficient completeness and exposure, not only three career aggregate rows. Standardize features to zero mean and unit variance before PCA. Fit two components for the dashboard; report explained variance, loadings, contributing features and sample coverage. Retain additional components in the analytical report if two dimensions lose substantial information.

“Service Firepower & Attack” and “Defensive Endurance & Rally Resilience” are proposed interpretive labels. Assign them only after inspecting loadings; component signs are arbitrary. Do not predetermine that Federer must score highest on PC1 or Nadal and Djokovic on PC2. PCA is dimensionality reduction; it does not itself assign clusters.

Comparison cohort and communication

An all-ATP archetype map requires an explicitly acquired and compatible broader cohort. If that extension is not feasible, label the map “Big Three charted playstyle observations.” Translate measures into plain language—first-try conversion, crisis recovery, and environmental versatility—while retaining the original tennis terms in tooltips.

Predictive modeling & validation

Target and prediction boundary

Core target: is_hold=1 if the serving player wins the completed service game, otherwise 0. The observation grain is one service game; tiebreaks are excluded or handled as a separately defined task. Reconstruct game boundaries and outcomes before modeling. The prediction moment is game start: use historical records dated before the match and legitimate pre-game context only.

Candidate inputs include server and returner identity, surface, match format, pre-match rolling serve/return rates, historical exposure, and pre-game set/game score. Same-game outcomes, final match winner, future statistics and post-match aggregates are forbidden. A separate match-win model would require a separate target, feature boundary and evaluation report.

Evaluation design

Use an untouched chronological test block with its date boundary selected after the sample audit and before tuning. Prefer five expanding-window time-aware validation splits within the earlier training block; keep all games from a match together. If five folds lack both classes, reduce fold count and document why. Stratified five-fold validation from the draft may be a secondary diagnostic only, with match grouping; it is not sufficient evidence of future performance.

Baseline, search and reporting

Compare a training-only surface hold-rate baseline and a simple logistic regression with XGBoost. Fit preprocessing and imputation inside validation folds. Optimize validation ROC-AUC while also reporting PR-AUC, log loss, Brier score and calibration. Inspect performance by player, surface, period and sample support. Do not promise a target ROC-AUC before measuring the data.

Parameter	Original candidate values
max_depth	3, 5, 7
learning_rate	0.01, 0.05, 0.1
n_estimators	50, 100, 200
subsample / colsample_bytree	Each: 0.8, 1.0

The complete grid has 108 combinations, or 540 fits across five folds before refitting. Start with a small pilot; if runtime exceeds the planned budget, use a documented staged or randomized search. Preserve random seeds, split IDs, model versions and search results. Freeze the final model before inspecting the test block.

Acceptance gate

Release model results only after verifying feature timestamps, disjoint match groups, training-only transformations, complete target reconstruction and a held-out comparison against baselines. If gains are negligible, publish that finding and prioritize interpretability and calibration.

Uncertainty, exports & simulator logic

Bootstrap protocol

Perform 1,000 resamples with replacement using a fixed seed. Resample matches as clusters so dependent points and games remain together. For held-out model metrics, resample held-out matches with their fixed predictions; this quantifies test-sample variability, not training uncertainty. Record the fraction of invalid resamples where a metric cannot be computed, such as ROC-AUC with a single class.

For descriptive player metrics, recompute count-based rates and composites within each resampled dataset. Preserve paired comparisons where the same match contributes to both players. Use the 2.5th and 97.5th percentiles for 95% intervals, and explicitly describe the population and dependence assumptions. An interval does not correct selection bias in charted coverage.

Tableau export contracts

File / grain	Required columns
tableau_goat_overview.csv Player × period × surface	Player; year/era; surface; matches; point exposures; metric estimates; uncertainty bounds; coverage indicators; feature version.
tableau_goat_spatial.csv Shot or aggregated zone	Match/player/shot identifiers or zone aggregate keys; shot type; surface; court side; coordinate provenance; analytical and render positions; count; jitter flag.
tableau_goat_predictions.csv Game or governed scenario	Game/scenario key; server; returner; surface; cutoff; observed outcome where available; hold probability; model/run version; evaluation context.

Offline simulator implementation

Keep the GOAT weight simulator separate from hold prediction. For continuously adjustable GOAT weights, export one record per player and bootstrap replicate containing every normalized component. Tableau calculates each replicate’s weighted score and summarizes its distribution for the selected weights. Do not calculate new composite bounds by simply weighting individual lower and upper bounds.

Alternatively, precompute a finite weight/scenario grid with estimated scores and intervals. In that mode, offer only supported discrete options and join by scenario_id; never imply arbitrary values have newly evaluated model predictions. If a matchup simulator is added, export its independently validated response grid and disclose interpolation or unsupported cases.

Reproducibility and publication controls

Every export includes an analysis-window label, data snapshot identifier and run/version key. Validate column types, unique grain, denominators, row counts and referential links. Use relational Tableau relationships to prevent match metrics multiplying when joined to shots. Aggregate large spatial exports if workbook responsiveness is poor.

Four-page dashboard experience

Design direction: “Swiss Precision Dark Mode.” Background #0D1117; gold #FFD700; Federer crimson #E63946; Nadal clay orange #E76F51; Djokovic accent #2A9D8F. The last hex is a teal-green accent; retain the exact color rather than relying on a “neon” description. Use high-contrast labels and redundant player names or symbols.

Page	Layout and interaction
1 • Overview & GOAT simulator	Hero profiles and era label; comparable KPI table with denominators; PCA map with loadings explanation; component-weight controls; weighted score and 95% interval; coverage and reset controls.
2 • Federer: The Maestro	Scoped header KPIs; serve placement and court-zone views; surface win rates; playstyle trends across eras; calculated tactical-change components; optional sourced biography panel.
3 • Nadal: The King of Clay	Scoped header KPIs; baseline depth and ad-court placement when supported; clay versus other-surface performance; era changes; confidence and coverage tooltips.
4 • Djokovic: The Machine	Scoped header KPIs; baseline direction and serve-zone views when supported; return and pressure profile; surface comparisons; era changes and coverage context.

Interaction specification

Global filters: year/era, surface and match format; page-local shot-type controls where relevant. Provide navigation buttons, selected-state labels, clear empty states and a reset action. Tooltips state metric definition, sample size, time window, coordinate provenance and any missing component. Parameter changes must visibly update scores and their supported uncertainty summaries.

Canvas and accessibility

Start with a fixed 1920 × 1080 desktop canvas and 10 px container padding as proposed. Review the actual published experience at common laptop widths; create a simplified layout if scaling makes labels unreadable. Maintain a consistent hierarchy and avoid communicating rank solely through color. Use plain-language legends and avoid decorative charts that obscure comparisons.

Content verification

The supplied lifetime weeks-at-No.1 and prize-money figures, brand deals, philanthropy details and asserted peak periods are not approved dashboard values. They require authoritative sources and explicit “as of” dates, or should be removed. Lifetime career context must be visually separated from 2010–2022 analytical KPIs. The proposal deliberately does not treat those figures as verified facts.

Delivery roadmap & acceptance gates

Planning estimate: 10 weeks at approximately 10–14 hours per week for one contributor. This is a relative schedule, not a committed calendar. Scope and effort should be revised after the first-week audit; preserve the data and evaluation gates before adding optional polish.

Week / checkpoint	Work package	Exit evidence
1 • I	Audit sources, licensing, coverage, headers, and target feasibility.	Source manifest; feasibility decision; frozen cohort rules.
2 • I	Chunked ingestion, adapters, SQLite schema and provenance.	Repeatable imports; valid joins; no duplicates/orphans.
3 • II	Code parser, null policy, chronology and zone transforms.	Quality report; manually reviewed point/shot fixtures.
4 • III	DR, clutch, SAI, era metrics and feature dictionary.	Hand-checked metric examples; denominators and fallback rules.
5 • IV	EDA, PCA and interpretable feature analysis.	Coverage plots; loadings; variance report; approved map labels.
6–7 • V	Game reconstruction, baselines, validation, tuning and bootstrap.	Timestamp audit; frozen model; test report; valid intervals.
8 • VI	Export contracts and four dashboard pages.	Reconciled totals; consistent filters; working parameters.
9 • VI–VII	Dashboard QA, usability review and reproducibility pass.	Published-view checks; fresh-run result; documented issues.
10 • VII	Documentation, presentation, final review and handoff.	Repository release; workbook; method report; final demo.

Dependencies and ownership

The project contributor owns data, implementation, analysis and dashboard delivery. An optional academic or domain reviewer checks tennis definitions, leakage boundaries and conclusions at weeks 1, 4 and 7. Modeling depends on validated games; spatial visuals depend on decoded zones; the simulator depends on normalized metrics and uncertainty exports.

Scope fallback order

First defer equipment and serve-speed innovation extensions; next defer a broader ATP cohort and biography panels. Preserve match comparisons, pressure/surface metrics, transparent PCA, a baseline-tested model where feasible, and four clear dashboard pages. If game labels cannot be reconstructed reliably, replace the predictive deliverable with a documented feasibility report or approve a separately scoped match-win task.

Quality assurance, risks & resources

Risk	Impact / mitigation
Incomplete charting coverage	Selection bias. Keep full-match versus charted samples separate; report coverage and avoid unsupported population claims.
Unsupported coordinates or speed	Misleading precision. Use categorical zones and omit unsupported variables.
Temporal / match leakage	Inflated performance. Audit timestamps, split by chronology, group by match, and fit transforms within folds.
Small surface/era samples	Unstable composites and intervals. Show exposures, suppress unsupported comparisons, and test sensitivity.
Unclear GOAT weights	Rankings depend on judgment. Publish normalization, allow weight exploration, and show rank stability.
Compute or workbook limits	Delayed delivery. Pilot search runtime, constrain parallelism, aggregate spatial outputs and test published responsiveness.
Mixed time windows or unsourced KPIs	Misleading narrative. Label 2010–2022 throughout; source and date separate lifetime context.

Verification plan

Data: test date boundaries, participant mapping, duplicate keys, referential integrity, supported surfaces and known parser examples.
Metrics: hand-check zero denominators, missing components, count aggregation, bounded rates and format splits.
Model: verify no match overlap, no future features, training-only imputation and appropriate baseline comparisons.
Spatial: verify orientation, court limits, zone assignment and separation of analytical from jittered fields.
Dashboard: reconcile KPIs to exports; test filter combinations, parameter weights, interval behavior, navigation and empty states.

Tools and budget assumptions

Use a local workstation, VS Code, Git, Python, pandas, NumPy, scikit-learn, XGBoost, SQLite, plotting libraries and Tableau Public. No paid API is assumed. Confirm current platform capabilities, publication constraints and package versions at kickoff. Pin dependencies; record seeds and hardware; exclude regenerable raw/model artifacts from Git where appropriate. Reserve 10–15% of effort for parser fixes and coverage issues within the overall estimate.

Definition of done

A clean environment can recreate validated exports from frozen inputs using documented commands. All four dashboard pages reconcile to those exports. Metrics and predictions carry source/window labels and sample support. Known limitations, license obligations, baseline comparisons, interval methods and optional omissions are documented; no placeholder performance scores remain in the published experience.

Implementation modules & handoff

Module	Responsibility / contract
src/ingestion.py	ingest_and_filter_match_charting_data(): source adapters, cohort filters, transactions and import counts.
src/cleaning.py	clean_matches(), parse_point_codes(), normalize_coordinates(): typed outputs, rejection logs and coordinate provenance.
src/features.py	compute_dominance_ratio(), compute_clutch_index(), compute_sai(), build_game_features(): counts, rates and historical cutoffs.
src/pca_clustering.py	run_pca_pipeline(): eligibility, scaling, component coordinates, loadings and explained variance; no implied clustering.
src/ml_model.py	train_xgboost(), evaluate_baselines(), run_bootstrapping(): grouped temporal splits, fitted models, test metrics and CIs.
src/export.py	export_tableau_extracts(): grain checks, schema contracts, simulator inputs and run metadata.
pipeline.py	Orchestrates stages with configuration, logging, validation gates and reproducible run identifiers.

Repository organization

tennis-goat-analytics-engine/
data/raw/ — frozen inputs or documented download manifests
data/processed/ — goat_master.db and derived feature tables
data/exports/ — the three governed Tableau CSVs
src/ — modular pipeline implementation
notebooks/exploratory_analysis.ipynb — exploratory analysis
tests/ — parser, metrics, integrity and leakage checks
reports/ — coverage, model evaluation and methodology
tableau/ — workbook and dashboard assets
pipeline.py • requirements.txt • README.md • config/

Handoff and presentation

The README explains setup, source acquisition, stage execution, output locations and known limitations. The feature dictionary defines grain, units, formulas, timestamps and exposure thresholds. The methods report explains sample coverage, PCA interpretation, model validation and bootstrap design. Deliver a short walkthrough that shows one analytical insight, one preference-sensitive ranking and one limitation.

Portfolio statement: use after completion

“Built a reproducible Python and SQLite tennis analytics pipeline for 2010–2022, engineered dominance and pressure metrics, evaluated PCA playstyle patterns and an XGBoost service-game model, and delivered a four-page Tableau Public dashboard with transparent coverage and uncertainty.” Adjust the wording to match completed work and measured results.

Assumptions, decisions & references

Decisions to confirm at kickoff

1. Approve a full ATP match-results source alongside the charting source for era-wide performance comparisons.
2. Confirm game-hold prediction as the supervised target, and select the held-out date boundary before tuning.
3. Decide whether the PCA map uses Big Three observations only or a broader compatible ATP cohort.
4. Approve zone-based spatial rendering wherever exact tracking coordinates are unavailable.
5. Lock normalization bounds and missing-component rules before calculating composites.
6. Choose continuous replicate-based GOAT weights or a finite scenario grid.
7. Confirm publication and redistribution terms, plus reviewer availability and weekly effort.

Changes from the supplied draft

The proposal retains the seven checkpoints, local architecture, cohort window, weighted innovation formula, PCA/XGBoost direction, 1,000 bootstrap iterations and four-page visual concept. It adds point/game tables, coordinate provenance, explicit exposure rules, feasibility gates and a delivery roadmap. It prioritizes chronological grouped evaluation over ordinary stratified folds and separates game-hold predictions from a preference-driven GOAT ranking. These refinements address implementation and interpretation risks rather than asserting completed findings.

Reference register

[1] Jeff Sackmann. **The Match Charting Project**, repository README, data files and data dictionary. Reviewed 29 September 2026. Repository: github.com/JeffSackmann/tennis_MatchChartingProject. Primary basis for the charting-source description; freeze and inspect the data dictionary during implementation.

[2] Jeff Sackmann. **tennis_atp**. Proposed complementary source for ATP match results and statistics. Repository: github.com/JeffSackmann/tennis_atp. Acquisition, field mapping and license review remain kickoff tasks; integration has not been performed.

[3] User-supplied **Pasted text.txt**, “Project Proposal & Technical Specification (BSc Data Science Level).” Basis for the project title, scope, checkpoints, formulas, proposed modules and dashboard design. Provided with this request.

Evidence status

This is a proposal, not a report of executed analysis. No ingestion counts, trained model results, computed innovation scores, published dashboard, or externally verified player biography statistics are claimed. Schedule, effort, component labels and optional features are planning assumptions subject to the documented gates.